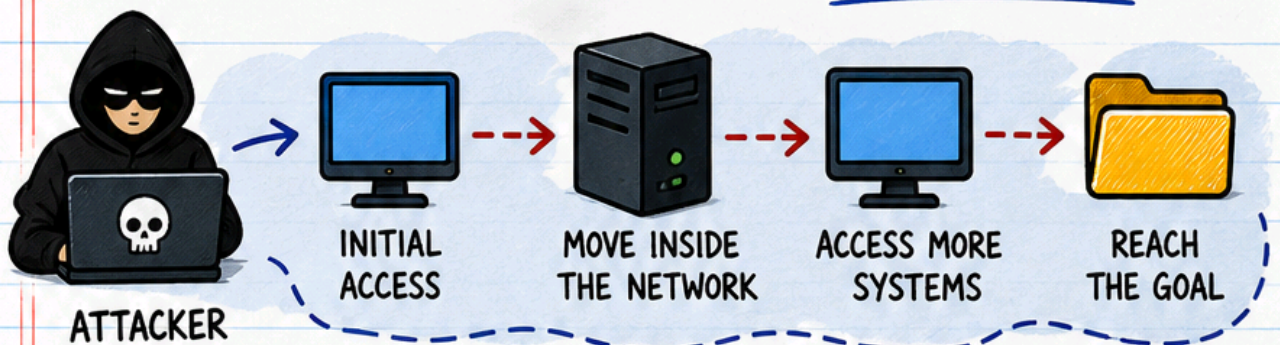


CYBERSECURITY NOTES

LATERAL MOVEMENT

DETECTION

DETECT EARLY. RESPOND FAST. STOP ATTACKERS.



UNUSUAL
AUTHENTICATION
PATTERNS



REMOTE
EXECUTION
TOOLS



ABNORMAL
NETWORK
TRAFFIC



PRIVILEGE
ESCALATION
INDICATORS

DETECTION CHECKLIST

- ✓ Monitor logs and alerts
- ✓ Look for unusual patterns
- ✓ Use the right tools (SIEM, IDS/IPS)
- ✓ Investigate every alert
- ✓ Respond quickly
- ✓ Review and improve

GOOD DETECTION
=
STRONG DEFENSE
=
SAFE NETWORK



LEARN



DETECT



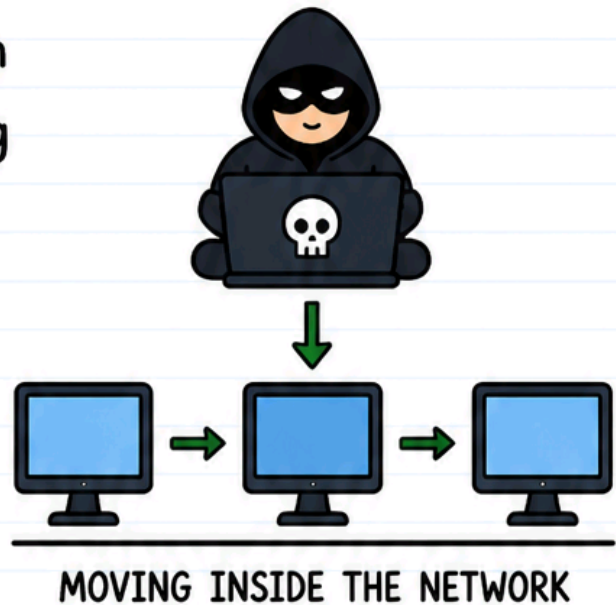
DEFEND



SUCCEED

1. WHAT IS LATERAL MOVEMENT?

Lateral movement is when an attacker, after getting inside one system, **moves to other systems** in the same network to find **more access, more data** and **more control**.



SIMPLE EXPLANATION



Think of your company network as a building. Attacker breaks in through one door (one computer). Then he walks through other rooms to open more doors and reach the most important room. That walking from one room to another is called **LATERAL MOVEMENT**.



WHY ATTACKERS DO THIS?

- ✓ To steal more data
- ✓ To get admin (full) access
- ✓ To stay hidden for a long time
- ✓ To take full control of the network



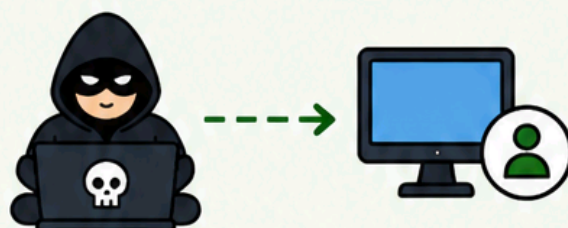
2. TYPES OF LATERAL MOVEMENT



After getting inside one system, attackers use different ways to move to other systems in the same network.

1 USER-BASED MOVEMENT

Using stolen or weak user accounts to login to other systems.



2 REMOTE SERVICES

Using remote tools and services like RDP, PsExec, WMI to access other systems.



3 PASS-THE-HASH

Using password hashes instead of passwords to login to other systems.



4 EXPLOITING VULNERABILITIES

Using weak or unpatched systems to exploit and get access to other systems.



5 FILE/SHARE ACCESS

Using shared folders, files or network drives to move to other systems.



Attackers use one or more of these methods to move deeper and deeper in the network.

3. HOW DOES LATERAL MOVEMENT HAPPEN? ≡



Attackers follow steps to move from one system to another inside the same network.

1 GET IN

Attacker enters the network using phishing, weak password or some exploit.



2 FIND SYSTEMS

Attacker looks for other computers, servers and users in the network.



3 STEAL CREDENTIALS

Attacker steals usernames, passwords or tokens from the system.



4 ACCESS OTHER SYSTEMS

Attacker uses stolen credentials or tools to login to other systems.



5 MOVE DEEPER

Attacker repeats the process to move from one system to another.



6 REACH THE GOAL

Finally, attacker reaches important servers, steals data or takes full control of the network.



Lateral movement = Attacker moving step by step from one system to another to get more access, more data and more control.

4. LATERAL MOVEMENT – CINEMATIC STORY EXAMPLE



Let's see a simple story of how an attacker moves inside a company network step by step.

1 THE ENTRY



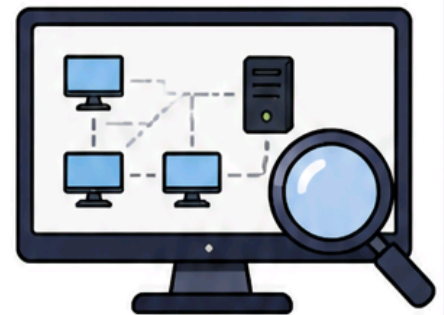
Attacker sends a fake email to an employee. The employee clicks the link and the attacker gets inside the network.

2 FIRST SYSTEM



Now attacker is inside Ravi's laptop which is connected to the company network.

3 LOOK AROUND



Attacker scans the network and finds other computers, servers, shared folders and users.

4 MOVE AHEAD



Using stolen credentials and tools, attacker moves from Ravi's laptop to another computer in the network.

5 REACH THE PRIZE



Finally attacker reaches the main server where important data, files and databases are stored.

6 THE DAMAGE



Attacker steals data, encrypts files or deletes important information. Big damage happens!



WHAT HAPPENED?

- Attacker did not break the company from outside again and again.
- He moved inside step by step from one system to another.
- This is called **LATERAL MOVEMENT**.



5. HOW AI HELPS ATTACKERS & DEFENDERS?



AI (Artificial Intelligence) is like a super smart helper. Both attackers and defenders use it.

HOW AI HELPS ATTACKERS



1. SMARTER PHISHING

AI writes very real fake emails that look exactly like real ones.



2. FIND WEAK SPOTS FAST

AI scans systems quickly and finds weak points to attack.



3. AUTOMATE ATTACKS

AI tools can run attacks automatically without human help.



4. BYPASS SECURITY

AI changes attack patterns to avoid detection by security systems.



5. LEARN AND IMPROVE

AI learns from the network and makes future attacks even better.

HOW AI HELPS DEFENDERS



1. DETECT THREATS FAST

AI can find strange activities and attacks in real-time.



2. ANALYZE BIG DATA

AI can read tons of logs and data very fast.



3. REDUCE FALSE ALERTS

AI helps to give only real alerts and ignore fake ones.



4. AUTOMATE RESPONSE

AI can block attacks and respond quickly without waiting for humans.



5. PREDICT FUTURE ATTACKS

AI can learn from past data and predict future attacks before they happen.



KEY TAKEAWAY



Attackers use AI to attack better and faster.

Defenders use AI to detect, stop and prevent attacks.

So, AI is a double-edged sword.





DETECTION STEP 1

UNUSUAL AUTHENTICATION PATTERNS



Attackers use stolen or weak accounts to login to other systems in the network. They try to act like normal users so that no one notices.

WHAT TO LOOK FOR?

Watch for login activities that look different from normal user behavior.



-  Login from a new or strange place  User is from India, but login happens from USA.
-  Login at a very strange time  User always logs in at 9 AM, but today logs in at 2 AM.
-  Login from a new device  User usually uses Windows laptop, but now login from an unknown device.
-  Too many failed login attempts  Many wrong passwords tried before a successful login.
-  Login to systems not used before  User who never uses a server suddenly logs in to it.

WHY IS IT IMPORTANT?



- This is often the **FIRST** sign of lateral movement.
- If we catch it early, we can **stop the attacker** before he moves further.

HOW TO DETECT?



- Use **SIEM** tools to monitor all login activities.
- Set **alerts** for unusual logins.
- Compare with user's **normal behavior**.

EXAMPLE



Ravi usually logs in from Office Laptop at 9 AM.



Today at **1:30 AM**, login happens from another country.



Using an **unknown device**.



This is **UNUSUAL!** Our system should alert the SOC team.



DETECTION STEP 2

REMOTE EXECUTION TOOLS



Attackers use remote tools to control other systems from far away. These tools help them run commands, steal data and move inside the network.

WHAT ARE REMOTE EXECUTION TOOLS?

These are programs or built-in features that allow someone to run commands on another computer.



EXAMPLES OF REMOTE EXECUTION TOOLS

PsExec



Used to run commands on remote Windows systems.

WMI



Windows Management Instrumentation used for remote management.

PowerShell



Powerful tool to run commands remotely on Windows systems.

Windows Remote Management (WinRM)



Microsoft service that allows remote management.

RDP



Remote Desktop Protocol used to control systems remotely.

WHAT TO LOOK FOR?



Unusual use of tools like PsExec, WMI, PowerShell or WinRM.



Commands executed on remote systems that are not normal.



Connections to many systems in a short time.



Use of admin accounts to run remote commands.



Creation of suspicious services or scheduled tasks.

HOW TO DETECT?



Monitor logs from Windows Event IDs like **4688**, **4624**, **4648**, **4776**, **7045**.



Use **SIEM** tools to detect unusual remote tool usage.



Look for unusual patterns in PowerShell or command executions.



Set alerts for multiple remote connections or admin logins.



Regularly review system and network logs.

EXAMPLE SCENARIO



Attacker gets inside the network.



Attacker uses PsExec to run command on another computer.



Command runs successfully on the remote system.



Attacker accesses files and steals important data.



We detect it and stop the attacker before more damage.



Remote execution tools are powerful for admins, but dangerous in wrong hands. Detect them early, stop attackers quickly!



DETECTION STEP 3

ABNORMAL NETWORK TRAFFIC



During lateral movement, attackers scan the network, connect to many systems and move data. This creates unusual or suspicious network traffic.

WHAT IS ABNORMAL NETWORK TRAFFIC?

It is any network activity that is different from normal daily network behavior. It can be a sign that an attacker is moving inside the network.



EXAMPLES OF ABNORMAL NETWORK TRAFFIC

1. PORT SCANNING



Attacker scans many ports on many systems to find open doors.

2. MULTIPLE SMB CONNECTIONS



Too many SMB connections between systems in a short time.

3. RDP CONNECTIONS SPIKE



Sudden increase in RDP connections to many systems.

4. LARGE DATA TRANSFER



Unusually large data being copied or moved between systems.

5. UNUSUAL PROTOCOL USAGE



Use of uncommon protocols or tools inside the network.

WHAT TO LOOK FOR?



Many connections to multiple systems from one source.



Connections happening very fast in a short time.



Access to admin shares like C\$, ADMIN\$ or IPC\$.



Unusual protocols or strange ports being used.



One user connecting to many systems that he should not access.

HOW TO DETECT?



Use **SIEM** tools to monitor firewall, proxy, DNS and network logs.



Look for spikes in traffic to ports like **445 (SMB), 3389 (RDP), 135, 139, 5985, 5986.**



Use **IDS/IPS** to detect port scanning and unusual protocols.



Create **alerts** for abnormal traffic patterns.



Regularly review network **baselines** and update rules.

EXAMPLE SCENARIO



Attacker is inside the network.



He scans many IPs and ports.



He connects to many systems (SMB/RDP).



He copies data from one system to another.



We detect unusual traffic and stop the attacker.



Abnormal network traffic is a big red flag for lateral movement. Detect it early, investigate and stop the attacker!



DETECTION STEP 4

PRIVILEGE ESCALATION INDICATORS



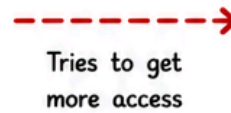
Attackers try to get higher privileges (more access) so they can control systems, change settings and avoid detection.
We need to detect these activities early.

WHAT IS PRIVILEGE ESCALATION?

It is when a user or attacker gets more access or permissions than they should have on a system or network.



Normal User
(Limited Access)



High Privilege User
(Full Access)

EXAMPLES OF PRIVILEGE ESCALATION INDICATORS

1 ADMIN GROUP ADDITION



User is added to admin groups like Domain Admins, Local Admins.

2 NEW ADMIN ACCOUNT



New admin accounts are created on systems.

3 SUDO / UAC BYPASS



Attacker tries to bypass sudo or UAC to get higher access.

4 SERVICE OR TASK CREATION



New services or scheduled tasks are created with high privileges.

5 TOKEN OR CREDENTIAL DUMP



Attackers dump credentials or tokens to get more privileges.

WHAT TO LOOK FOR?



Users suddenly becoming admin.



New admin accounts that are not normal.



Use of tools like Mimikatz, PowerUp, BloodHound, PsExec.



New services, tasks or drivers installed.



Access to sensitive files like SAM, SYSTEM, LSASS memory.

HOW TO DETECT?



Use **SIEM** to monitor logs like 4624, 4672, 4688, 4720, 4728, 4732.



Monitor changes in admin groups and user privileges.



Look for unusual use of privilege escalation tools.



Set alerts for new admin accounts, services and scheduled tasks.



Review endpoint and server logs regularly.

EXAMPLE SCENARIO



Attacker is inside the network.



He adds himself to "Domain Admins" group.



He creates a new service to stay hidden.



He dumps credentials to get more access on other systems.



We detect it and stop the attacker before damage.



KEY TAKEAWAY: Privilege escalation gives attackers more power.
Detect it early, stop it quickly and protect the network.



FINAL SUMMARY

LATERAL MOVEMENT DETECTION



Lateral Movement is how attackers move inside a network from one system to another. Early detection helps us stop them before they cause big damage.

4 KEY DETECTION STEPS

1

UNUSUAL AUTHENTICATION PATTERNS



Detect strange logins like new locations, odd times, unknown devices or too many failed attempts.

2

REMOTE EXECUTION TOOLS



Watch for use of tools like PsExec, WMI, PowerShell, WinRM, RDP that help attackers run commands remotely.

3

ABNORMAL NETWORK TRAFFIC



Look for port scanning, many SMB/RDP connections, large data transfer or strange protocols.

4

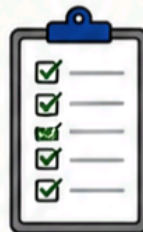
PRIVILEGE ESCALATION INDICATORS



Detect actions that give more privileges like new admin accounts, group changes, token dumps or suspicious services.

DETECTION CHECKLIST

- ✓ Monitor all login activities.
- ✓ Check for unusual times, locations or devices.
- ✓ Watch for remote execution tool usage.
- ✓ Monitor network traffic for unusual patterns.
- ✓ Look for multiple connections in short time.
- ✓ Check for admin or privilege changes.
- ✓ Monitor creation of new accounts or services.
- ✓ Use SIEM and IDS/IPS for alerts.
- ✓ Review logs regularly.
- ✓ Respond quickly and stop the attacker.



WHAT ATTACKERS TRY TO DO



- Get inside the network.
- Move from one system to another.
- Steal data and keep control.
- Hide their activities.
- Get higher privileges.

WHAT WE SHOULD DO



- Detect early.
- Understand attacker behavior.
- Use the right tools and alerts.
- Investigate every alert.
- Stop the attack and secure the network.



KEY TAKEAWAY



Lateral movement is a silent and dangerous phase of an attack.
Detect early. Respond fast. Stop attackers before they reach their goal.

GOOD DETECTION = STRONG DEFENSE = SAFE NETWORK



LATERAL MOVEMENT DETECTION - COMPLETE SUMMARY

★ Congratulations! You have completed your notes on **LATERAL MOVEMENT DETECTION**. ★

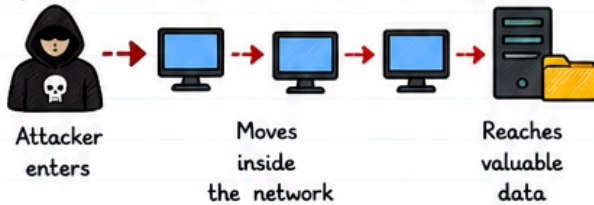


Lateral Movement is how attackers move inside a network from one system to another to reach their final goal.
Detect it early. Stop it quickly. Protect everything!



WHAT IS LATERAL MOVEMENT?

Attackers move from one system to another inside the network to find valuable data, gain more access and reach their target.



WHY DETECT LATERAL MOVEMENT?

- ★ Attackers hide and move silently.
- ★ It helps them avoid detection.
- ★ They steal data and escalate privileges.
- ★ Early detection stops big attacks.
- ★ Protects users, data and the entire organization.



4 KEY DETECTION STEPS

1

UNUSUAL AUTHENTICATION PATTERNS



Detect strange logins like new locations, odd times, unknown devices or too many failed attempts.

2

REMOTE EXECUTION TOOLS



Watch for tools like PsExec, WMI, PowerShell, WinRM, RDP used to run commands remotely.

3

ABNORMAL NETWORK TRAFFIC



Look for port scanning, many SMB/RDP connections, large data transfer or unusual protocols.

4

PRIVILEGE ESCALATION INDICATORS



Detect actions that give more privileges like new admin accounts, group changes, token dumps or suspicious services.

IMPORTANT DETECTION CHECKLIST

- ✓ Monitor all login activities.
- ✓ Check for unusual times, locations or devices.
- ✓ Watch for remote execution tool usage.
- ✓ Monitor network traffic for unusual patterns.
- ✓ Look for multiple connections in a short time.
- ✓ Check for admin or privilege changes.
- ✓ Monitor creation of new accounts or services.
- ✓ Use SIEM and IDS/IPS for alerts.
- ✓ Review logs regularly.
- ✓ Respond quickly and stop the attacker.



WHAT ATTACKERS TRY TO DO



- Get inside the network.
- Move from one system to another.
- Steal data and keep control.
- Hide their activities.
- Get higher privileges.

WHAT WE SHOULD DO



- Detect early.
- Understand attacker behavior.
- Use the right tools and alerts.
- Investigate every alert.
- Stop the attack and secure the network.

YOUR NEXT STEPS

- 1 Practice daily in labs and real scenarios.
- 2 Learn new tools, techniques and attack trends.
- 3 Document your investigations and findings.
- 4 Stay updated with threat intelligence.
- 5 Build your portfolio and help others.



CONTINUE YOUR JOURNEY WITH **BLUETEAMERS.IO**

- ✓ Hands-on SOC Training
- ✓ Real-World Case Studies
- ✓ Practice Labs & Simulations
- ✓ Beginner to Advanced Level
- ✓ Career Guidance & Mentorship
- ✓ Active Community Support

Blueteamers.io is built to help you become a confident, job-ready SOC Analyst!



Better you understand attacks, better you can defend them.
Keep Learning. Keep Growing. Keep Defending! 😊



Thank You for completing these notes! You are Awesome!

